

Multiple Choice (10 marks)

1. In Python 3, what is the output of `print tuple[0]` if `tuple = ( 'abcd', 786 , 2.23, 'john', 70.2 )`?
 - (a) ('abcd', 786 , 2.23, 'john', 70.2)
 - (b) abcd
 - (c) Error
 - (d) None of the above.
2. Which types of functions grow the slowest?
 - (a) $O(\log e^N)$
 - (b) $O(\log N)$
 - (c) $O(\log \log N)$
 - (d) $O(N)$
3. What is the value of this python expression: `4 + 3 % 2`?
 - (a) 1
 - (b) 6
 - (c) 4
 - (d) 5
4. Let `l = [1,2,3,4]`. What is `max(l)` in Python 3?
 - (a) 1
 - (b) 2
 - (c) 3
 - (d) 4
5. The boolean expression `a[i] == max || !(max != a[i])` can be simplified to
 - (a) `a[i] == max`
 - (b) `a[i] != max`
 - (c) `a[i] < max || a[i] > max`
 - (d) FALSE

True / False (10 marks)

Answer True or False

6. True or False: The output of the expression `'abc'[::-1]` in Python 3 is `'cba'`, which reverses the string.
7. True or False: The output of `set(1)` in Python 3 will remove duplicates and result in `{1, 2, 3, 4}`.
8. True or False: In Python, the expression `4 * 1 ** 4` evaluates to 4.
9. True or False: In Python 3, the operator for floor division is represented by the single slash (`/`).
10. True or False: In Python 3, using negative indexing on a string always results in an error.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Explain how the RGB color model represents colors using red, green, and blue values. Discuss how many bits are required to represent a color in this model and why.
12. Discuss how providing laptops or tablets to all students can positively influence the digital divide in education. What are the potential benefits and challenges of this policy?

End of Paper